

Question number	Answer	Additional guidance	Mark
4(a)(i)	• D is the correct answer A is not the correct answer as the ratio is incorrect B is not the correct answer as the ratio is incorrect as it is rounded incorrectly C is not the correct answer as the ratio is incorrect as it is rounded incorrectly		(1)

Question number	Answer	Additional guidance	Mark
4(a)(ii)	An answer that includes four of the following points - thermoreceptors produce action potentials below 42 (°C) / no action potentials for nociceptors below 42(°C) (1) <i>accept nociceptor produces action potentials after 42(°C)</i> - the number of action potentials from thermoreceptors does not increase after 44(°C) (1) - thermoreceptors produce action potentials over a greater range of temperatures than nociceptors (1) - nociceptors have larger increase in number of impulses for each degree rise in temperature / are more sensitive to a change in temperature (1) - comment on information missing on methodology (1)	accept nociceptors {produce more action potentials / more sensitive} at higher temperatures / above 45(.6)(°C) accept levels off / flattens out / peaks accept 34(.2)-50(°C) for thermoreceptor 42 – 50(°C) for nociceptor accept graph for nociceptors is steeper greater increase in number of action potentials for nociceptors e.g. size / age / area of skin / no sample size / no statistical analysis / no mean calculated accept comment on no error bars so no comment on significance	(4)

Question number	Answer	Additional guidance	Mark
4(b)	An explanation that includes four of the following points: • nicotine causes an increase in the number of action potentials (in post synaptic neurone) (1) • as nicotine binds to acetylcholine receptors causing action potential (1) • Groups B and C have similar number of action potentials (1) • if nicotine given {to mice with gene for acetylcholine receptor deleted / group C} then number of action potentials stays same (1) • as {no / less} acetylcholine receptor (on post synaptic membrane) (1)	accept if nicotine is not given the number of action potentials stays the same group A has the highest number of action potentials accept nicotine has a similar structure to acetylcholine ignore nicotine blocks receptors accept fewer action potentials in group C ignore no action potentials generated	(4)

